THE LARGEST OPEN WHERE A TAU-LOCAL PROPERTY HOLDS FOR THE STABILIZER IS R-INVARIANT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- tau-local properties and the largest open. Descent on Spaces, lemma largest-open-of-the-base (cross-chapter, named). Gives W with its pullback description.
- Stabilizer conjugation action. the stabilizer group space G -> U of this chapter (same chapter, structural anchor). Conjugation moves stabilizer points along groupoid arrows.

Source reference. Original section 020B: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 262–262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06R2: focused-seeds/geometry/S9-smg-property-G-invariant/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 044F: focused-seeds/geometry/S9-smg-property-G-invariant/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $B \to S$ be as in Section 020B. Let (U, R, s, t, c) be a groupoid in algebraic spaces over B. Let $G \to U$ be its stabilizer group algebraic space. Let $\tau \in \{fppf, étale, smooth, syntomic\}$. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which is τ-local on the target. Assume $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology. Let $W \subset U$ be the maximal open subspace such that $G_W \to W$ has property $\mathcal{P}$. Then W is R-invariant (see Groupoids in Spaces, Definition 044F).*

Proof. The existence and properties of the open $W \subset U$ are described in Descent on Spaces, Lemma 06R2. The morphism

$$G \times_{U,t} R \longrightarrow R \times_{s,U} G, \quad (g, r) \longmapsto (r, r^{-1} \circ g \circ r)$$

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is an isomorphism of algebraic spaces over R (where $\circ$ denotes composition in the groupoid). Hence $s^{-1}(W) = t^{-1}(W)$ by the properties of W proved in the aforementioned Descent on Spaces, Lemma 06R2. $\square$